

Chapter 3

Simulation of Shock Triggered Star Formation

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The physical process that causes the shock to compress and increase its density, thereby increasing the temperature according to laws of thermodynamics, is governed by the Jeans Mass (M_J) [4]. It basically tells you for a shock to hit and collapse a gas of cloud, the shock needs to do two things: first is to increase density creating a place for gravitational collapse to take place, the other is for the gas to cool sufficiently for the collapse to happen initially, without cooling, the gas cannot collapse into a protostar. The governing physical equation is:

$$M_J = \left(\frac{5k_B T}{G \mu m_H} \right)^{3/2} \left(\frac{3}{4\pi \rho} \right)^{1/2} \quad (3.1)$$

where k_B is the Boltzmann constant, T is the gas temperature, G is the gravitational constant, μm_H is the mean molecular weight, and ρ is the mass density. For studying shock dynamics, this simplifies to the fundamental proportionality:

$$M_J \propto \frac{T^{3/2}}{\rho^{1/2}} \quad (3.2)$$

To trigger star formation, a shockwave must increase ρ at a higher proportional rate than T^3 .

3.1 Phase I: The Planar Shock Approximation

3.1.1 The Sod Shock Tube Formulation

We initially tried to create a shock directly from Sedov solution of a Supernova (SNe) blastwave, but due to the explosion being a point source the simulation immediately crashed due to dt approaching zero very quickly. We implement a simpler model first for testing, we use a planar shock. The computational grid was divided into high-pressure driver gas and low-pressure ambient medium.

When the supersonic shock passes through the gas, the physics are governed by the Rankine-Hugoniot Jump Conditions [5]. For a shock moving at Mach number $\mathcal{M}$ into an ambient gas with adiabatic index γ , the density (ρ_2) and pressure (P_2) of the compressed gas are:

$$\frac{\rho_2}{\rho_1} = \frac{(\gamma + 1)\mathcal{M}^2}{(\gamma - 1)\mathcal{M}^2 + 2} \quad (3.3)$$

$$\frac{P_2}{P_1} = \frac{2\gamma\mathcal{M}^2 - (\gamma - 1)}{\gamma + 1} \quad (3.4)$$

The temperature jump, derived from the ideal gas law, is:

$$\frac{T_2}{T_1} = \frac{P_2 \rho_1}{P_1 \rho_2} \quad (3.5)$$

To Physically inject a cloud in our simulation we need strict geometric limit in the cartesian boundary. The cloud is modeled as a circular density enhancement in equilibrium with the surrounding Interstellar Medium (ISM).

The physical density field $\rho(x, y)$ applied to the grid is defined mathematically as a piecewise step-function, dependent on the radial distance from the cloud's center coordinates (x_c, y_c) :

$$\rho(x, y) = \begin{cases} \rho_{\text{cloud}} & \text{if } \sqrt{(x - x_c)^2 + (y - y_c)^2} \leq R_{\text{cloud}} \\ \rho_{\text{ambient}} & \text{otherwise} \end{cases} \quad (3.6)$$

where R_{cloud} is the defined radius of the molecular cloud.

We need to make sure that the injected cloud does not expand or collapse under its own internal pressure, that is, it must be stable, this will help us check if the collapse is just happening from the shock and not its internal instability. Therefore, the pressure field $P(x, y)$ must remain strictly uniform across the entire computational domain:

$$P_{\text{cloud}} = P_{\text{ambient}} \quad (3.7)$$

Within the FLASH `SN_CloudCrush` setup, which is the setup we created in FLASH to model this shock, this exact geometry and a stable density contrast of $\chi = 100$ were achieved by injecting the following physical parameters into the `flash.par` initialization file:

- `sim_rhoAmbient = 1.0` (Defines the background ISM density)
- `sim_pAmbient = 1.0` (Defines the background ISM pressure)
- `sim_rhoCloud = 100.0` (Injects the cloud mass, establishing $\chi = 100$)
- `sim_pCloud = 1.0` (Matches the ambient pressure to enforce thermodynamic equilibrium)

3.1.2 FLASH Implementation

To generate a Mach 2 shock in an ideal gas ($\gamma = 5/3$), the exact analytical ratios are $\rho_2/\rho_1 = 2.28$ and $P_2/P_1 = 4.75$. These were injected into `FLASH.par` file using the following parameters:

- `gamma = 1.6666667`
- `sim_rhoLeft = 2.28`
- `sim_pLeft = 4.75`
- `sim_uLeft = 1.45`

3.1.3 Results from planar shock

The simulated planar shock has the following density profile as it travels to the injected cloud

This shock while travelling through the cloud caused the collapse of the gas inside the stable cloud, as seen from the reduction of jean's mass from the plot below:

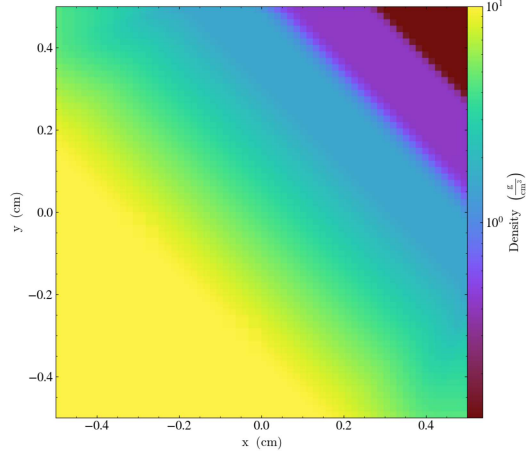


Figure 3.1: Profile of Planar shock

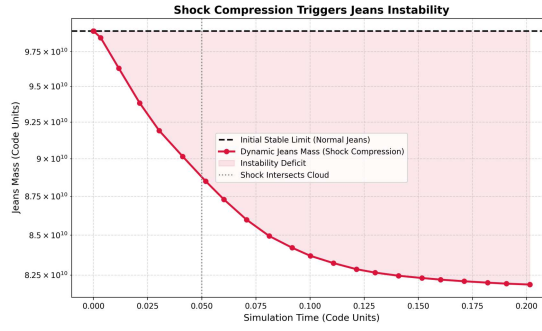


Figure 3.2: Observed reduction of Jean's mass due to shock

3.2 Phase II: The Supernova Shockwave and Cloud Compression

3.2.1 The Sedov-Taylor Blast Wave

To accurately model an SNe shock travelling through a cloud medium of gas, we use spherical explosion, A SNe in the energy-conserving Sedov-Taylor phase [6, 7] is governed by the energy of the blast (E) and the density of the surrounding interstellar

medium (ρ_{ISM}). The shock radius over time is given by:

$$R(t) = \xi_0 \left(\frac{E}{\rho_{ISM}} \right)^{1/5} t^{2/5} \quad (3.8)$$

where $\xi_0 \approx 1.15$. The velocity of the blast wave (v_s) decays over time:

$$v_s = \frac{2}{5} \xi_0 \left(\frac{E}{\rho_{ISM}} \right)^{1/5} t^{-3/5} = \frac{2R}{5t} \quad (3.9)$$

3.2.2 The Transmitted Shock and Triggered Collapse

When the supernova shock (v_s) hits the dense molecular cloud (ρ_c), it generates a transmitted shock that moves into the cloud [8]. Letting the density contrast be $\chi = \rho_c / \rho_{ISM}$ (where $\chi = 100$ in our setup), the velocity of the transmitted shock is:

$$v_{tc} \approx v_s \chi^{-1/2} \quad (3.10)$$

If the gas is adiabatic ($\gamma = 5/3$), the maximum compression is $4\times$, but the temperature scales as v_{tc}^2 . Per Equation 3.2, the $T^{3/2}$ term explodes, stabilizing the cloud. Therefore, we modeled the gas as nearly isothermal ($\gamma = 1.01$) to simulate perfect radiative cooling. For an isothermal shock, compression scales as the square of the Mach number ($\rho_2 / \rho_1 = \mathcal{M}^2$). Substituting this into the Jeans equation yields:

$$M_{J,2} = M_{J,1} \left(\frac{1}{\mathcal{M}^2} \right)^{1/2} = \frac{M_{J,1}}{\mathcal{M}} \quad (3.11)$$

This mathematically proves that an isothermal shock reduces the critical mass required for collapse exactly in inverse proportion to the Mach number.

3.2.3 Computational Parameters for Stability

To force the hydrodynamics engine to survive the extreme non-linearities of an isothermal shock hitting a density wall- the cloud, the following ‘‘Strong Shock’’ parameters were injected into `flash.par`:

- `sim_rhoCloud` = 100.0 (Density contrast $\chi = 100$)
- `sim_expEnergy` = 10.0 (Supernova energy)
- `sim_rInit` = 0.05 (Initial explosion volume to prevent point-singularity)

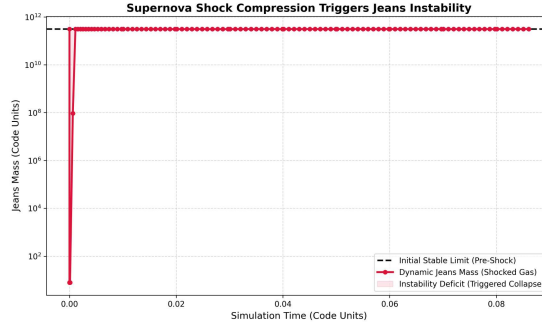


Figure 3.3: Jeans mass change as the shock hits the cloud

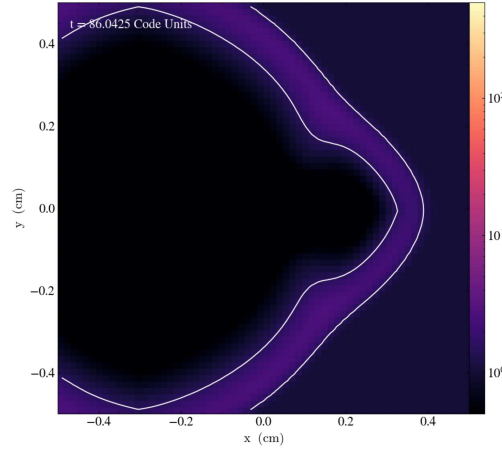


Figure 3.4: Shock front hitting the cloud and deforming as it hits the gas

- $\gamma = 1.01$ (Isothermal/Radiative cooling simulation)
- `RiemannSolver = "hllc"` (Prevents carbuncle instability)
- `interp_order = 1` (Prevents negative density overshoots)
- `shockDetect = .true.` and `use_flattening = .true.`

3.2.4 Results from SNe shock injection hitting the cloud

The change in jean's mass is plotted below:

The plot of shock hitting the cloud is given below:

The problem with this is we were not able to inject proper cooling model, that is why the jean's mass is seeminlgy decreasing in an instant then blowing up to the max mass required, In order to properly model the gas hitting the cloud, we need to define a proper model for cooling, if the gas does not have time to cool, it will never collapse.